Comparative metagenomic analysis from Sundarbans ecosystems advances our understanding of microbial communities and their functional roles.

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The Sundarbans mangrove, located at the mouth of the Ganges and Brahmaputra Rivers, is the world's largest tidal mangrove forest. These mangroves are also one of the most striking sources of microbial diversity, essential in productivity, conservation, nutrient cycling, and rehabilitation. Hence, the main objective of this study was to use metagenome analysis and provide detailed insight into microbial communities and their functional roles in the Sundarbans mangrove ecosystem. A comparative analysis was also done with a non-mangrove region of the Sundarbans ecosystem to assess the capability of the environmental parameters to explain the variation in microbial community composition. The study found several dominant bacteria, viz., Alphaproteobacteria, Actinomycetota, Bacilli, Clostridia, Desulfobacterota, Gammaproteobacteria, and Nitrospira, from the mangrove region. The mangrove sampling site reports several salt-tolerant bacteria like Alkalibacillus haloalkaliphilus, Halomonas anticariensis, and Salinivibrio socompensis. We found some probiotic species, viz., Bacillus clausii, Lactobacillus curvatus, Vibrio mediterranei and Vibrio fluvialis, from the Sundarbans mangrove. Nitrifying bacteria in Sundarbans soils were Nitrococcus mobilis, Nitrosococcus oceani, Nitrosomonas halophila, Nitrospirade fluvii, and others. Methanogenic archaea, viz., Methanoculleus marisnigri, Methanobrevibacter gottschalkii, and Methanolacinia petrolearia, were highly abundant in the mangroves as compared to the non-mangrove soils. The identified methanotrophic bacterial species, viz., Methylobacter tundripaludum, Methylococcus capsulatus, Methylophaga thiooxydans, and Methylosarcina lacus are expected to play a significant role in the degradation of methane in mangrove soil. Among the bioremediation bacterial species identified, Pseudomonas alcaligenes, Pseudomonas mendocina, Paracoccus denitrificans, and Shewanella putrefaciens play a significant role in the remediation of environmental pollution. Overall, our study shows for the first time that the Sundarbans, the largest mangrove ecosystem in the world, has a wide range of methanogenic archaea, methanotrophs, pathogenic, salt-tolerant, probiotic, nitrifying, and bioremediation bacteria.

Evaluation of synbiotic combinations of commercial probiotic strains with different prebiotics in in vitro and ex vivo human gut microcosm model.

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Exploring probiotics for their crosstalk with the host microbiome through the fermentation of non-digestible dietary fibers (prebiotics) for their potential metabolic end-products, particularly short-chain fatty acids (SCFAs), is important for understanding the endogenous host-gut microbe interaction. This study was aimed at a systematic comparison of commercially available probiotics to understand their synergistic role with specific prebiotics in SCFAs production both in vitro and in the ex vivo gut microcosm model. Probiotic strains isolated from pharmacy products including Lactobacillus sporogenes (strain not labeled), Lactobacillus rhamnosus GG (ATCC53103), Streptococcus faecalis (T-110 JPC), Bacillus mesentericus (TO-AJPC), Bacillus clausii (SIN) and Saccharomyces boulardii (CNCM I-745) were assessed for their probiotic traits including survival, antibiotic susceptibility, and antibacterial activity against pathogenic strains. Our results showed that the microorganisms under study had strain-specific abilities to persist in human gastrointestinal conditions and varied anti-infective efficacy and antibiotic susceptibility. The probiotic strains displayed variation in the utilization of six different prebiotic substrates for their growth under aerobic and anaerobic conditions. Their prebiotic scores (PS) revealed which were the most suitable prebiotic carbohydrates for the growth of each strain and suggested xylooligosaccharide (XOS) was the poorest utilized among all. HPLC analysis revealed a versatile pattern of SCFAs produced as end-products of prebiotic fermentation by the strains which was influenced by growth conditions. Selected synbiotic (prebiotic and probiotic) combinations showing high PS and high total SCFAs production were tested in an ex vivo human gut microcosm model. Interestingly, significantly higher butyrate and propionate production was found only when synbiotics were applied as against when individual probiotic or prebiotics were applied alone. qRT-PCR analysis with specific primers showed that there was a significant increase in the abundance of lactobacilli and bifidobacteria with synbiotic blends compared to pre-, or probiotics alone. In conclusion, this work presents findings to suggest prebiotic combinations with different well-established probiotic strains that may be useful for developing effective synbiotic blends.

Bacillus subtilis SF106 and Bacillus clausii SF174 spores reduce the inflammation and modulate the gut microbiota in a colitis model.

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Chronic intestinal inflammation is associated with strong alterations of the microbial composition of the gut. Probiotic treatments and microbiota-targeting approaches have been considered to reduce the inflammation, improve both gut barrier function as well as overall gastrointestinal health. Here, a murine model of experimental colitis was used to assess the beneficial health effects of Bacillus subtilis SF106 and Bacillus clausii (recently renamed Shouchella clausii) SF174, two spore-forming strains previously characterised in vitro as potential probiotics. Experimental colitis was induced in BALB/c mice by the oral administration of dextran sodium sulphate (DSS) and groups of animals treated with spores of either strain. Spores of both strains reduced the DSS-induced inflammation with spores of B. clausii SF174 more effective than B. subtilis SF106. Spores of both strains remodelled the mouse gut microbiota favouring the presence of beneficial microbes such as members of the Bacteroidetes and Akkermansia genera.

Bacillus clausii Bacteremia Following Probiotic Use: A Report of Two Cases.

PMID: 38721227 | Indexed in PubMed

The use of probiotics to improve bacterial flora and achieve control of diarrheal episodes is a common practice in outpatients and hospitalized patients. In most cases, related adverse events are few and not life-threatening. However, cases of bacteremia associated with the use of these substances have been described, mainly in the pediatric population in which their prescription is more common. Cases of bacteremia and sepsis have also been documented in immunocompetent and immunocompromised adult patients following the use of probiotics. We present the report of two patients who, in the context of diarrhea, received probiotics with Bacillus clausii spores during their stay in the intensive care unit. They subsequently developed sepsis and blood-culture-documented bacteremia. Both patients were treated with daptomycin as the final treatment regimen.

Promising clinical and immunological efficacy of Bacillus clausii spore probiotics for

supportive treatment of persistent diarrhea in children.

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Persistent diarrhea is a severe gastroenteric disease with relatively high risk of pediatric mortality in developing countries. We conducted a randomized, double-blind, controlled clinical trial to evaluate the efficacy of liquid-form Bacillus clausii spore probiotics (LiveSpo CLAUSY; 2 billion CFU/5 mL ampoule) at high dosages of 4-6 ampoules a day in supporting treatment of children with persistent diarrhea. Our findings showed that B. clausii spores significantly improved treatment outcomes, resulting in a 2-day shorter recovery period ($p < 0.05$) and a 1.5-1.6 folds greater efficacy in reducing diarrhea symptoms, such as high frequency of bowel movement of ≥ 3 stools a day, presence of fecal mucus, and diapered infant stool scale types 4-5B. LiveSpo CLAUSY supportive treatment achieved 3 days ($p < 0.0001$) faster recovery from diarrhea disease, with 1.6-fold improved treatment efficacy. At day 5 of treatment, a significant decrease in blood levels of pro-inflammatory cytokines TNF-α, IL-17, and IL-23 by 3.24% ($p = 0.0409$), 29.76% ($p = 0.0001$), and 10.87% ($p = 0.0036$), respectively, was observed in the Clausy group. Simultaneously, there was a significant 37.97% decrease ($p = 0.0326$) in the excreted IgA in stool at day 5 in the Clausy group. Overall, the clinical study demonstrates the efficacy of B. clausii spores (LiveSpo CLAUSY) as an effective symptomatic treatment and immunomodulatory agent for persistent diarrhea in children.Trial registration: NCT05812820.

Chromosomal aadD2 encodes an aminoglycoside nucleotidyltransferase in Bacillus clausii.

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Bacillus clausii SIN is one of the four strains of B. clausii composing a probiotic administered to humans for the prevention of gastrointestinal side effects due to oral antibiotic therapy. The strain is resistant to kanamycin, tobramycin, and amikacin. A gene conferring aminoglycoside resistance was cloned into Escherichia coli and sequenced. The gene, called aadD2, encoding a putative 246-amino acid protein, shared 47% identity with ant(4')-Ia from Staphylococcus aureus, which encodes an aminoglycoside 4'-O-nucleotidyltransferase. Phosphocellulose paper-binding assays indicated that the gene product was responsible for nucleotidylation of kanamycin, tobramycin, and amikacin. The aadD2 gene was detected by DNA-DNA hybridization in the three other strains of the probiotic mixture and in the reference strain B. clausii DSM8716, although it did not confer resistance in these strains. Mutations in the sequence of the putative promoter for aadD2 from B. clausii SIN resulted in higher identity with consensus promoter sequences and may account for aminoglycoside resistance in that strain. The aadD2 gene was chromosomally located in all strains and was not transferable by conjugation. These data indicate that chromosomal aadD2 is specific to B. clausii.

Molecular and biochemical characterization of the chromosome-encoded class A beta-lactamase BCL-1 from Bacillus clausii.

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A chromosomal beta-lactamase gene from Bacillus clausii NR, which is used as a probiotic, was cloned and expressed in Escherichia coli. It encodes a clavulanic acid-susceptible Ambler class A beta-lactamase, BCL-1, with a pI of 5.5 and a molecular mass of ca. 32 kDa. It shares 91% and 62% amino acid identity with the chromosomally encoded PenP penicillinases from B. clausii KSM-K16 and Bacillus licheniformis, respectively. The hydrolytic profile of this beta-lactamase includes penicillins, narrow-spectrum cephalosporins, and cefpirome. This chromosome-encoded enzyme was inducible in B. clausii, and its gene is likely related to upstream-located regulatory genes that share significant identity with those reported to be upstream of the penicillinase gene of B. licheniformis. The bla(BCL-1) gene was located next to the known chromosomal aadD2 gene and the erm34 gene, which encode resistance to aminoglycosides and macrolides, respectively. Similar genes were found in a collection of B. clausii reference strains.

Bacteremia after Bacillus clausii administration for the treatment of acute diarrhea: A case report.

PMID: 34669274 | Indexed in PubMed

Bacillus clausii is a gram-positive rod used as a probiotic to treat diarrhea and the side effects of antibiotics such as pseudomembranous colitis. We report a case of B. clausii bacteremia in a non-immunocompromised patient with active peptic ulcer disease and acute diarrhea. The probiotic was administered during the patient´s hospitalization due to diarrhea of infectious origin. B. clausii was identified in the bloodstream of the patient through Matrix-Assisted Laser Desorption Ionization-Time of Flight (MALDI-TOF) days after her discharge. Given the wide use of probiotics, we alert clinicians to consider this microorganism as a causative agent when signs of systemic infection, metabolic compromise, and hemodynamic instability establish after its administration and no pathogens have been identified that could explain the clinical course.

Comparative accounts of probiotic properties of spore and vegetative cells of Bacillus clausii UBBC07 and in silico analysis of probiotic function.

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In this study, the spores and vegetative cells of B. clausii were independently evaluated for probiotic properties such as acid, gastric juice, bile, and intestinal fluid tolerance, adhesion to solvents/mucin and zeta potential. In addition, in silico identification of genome features contributing to probiotic properties were investigated. The results showed that spores were highly stable at gastric acidity and capable to germinate and multiply under intestinal conditions as compared to vegetative cells. The higher hydrophobicity of spores, compared to vegetative cells, is advantageous for colonization and persistence in the intestine. Furthermore, the presence of F 0 F 1 ATP synthase, amino acid decarboxylase, bile acid symporter, mucin/collagen/fibronectin-binding proteins, heat/cold shock proteins, and universal stress proteins suggests that the strain is able to survive stress. In conclusion, the results demonstrate that B. clausii UBBC07 spores show significantly higher survival and adhesion in in vitro gastrointestinal conditions as compared to vegetative cells. Besides, this study provides a comparative analysis of the in vitro probiotic properties of spores and vegetative cells of Bacillus clausii UBBC07.

Comparative secretome analysis of four isogenic Bacillus clausii probiotic strains.

PMID: 23816335 | Indexed in PubMed

The spore-bearing alkaliphilic Bacillus species constitute a large, heterogeneous group of microorganisms, important for their ability to produce enzymes, antibodies and metabolites of potential medical use. Some Bacillus species are currently being used for manufacturing probiotic products consisting of bacterial spores, exhibiting specific features (colonization, immune-stimulation and antimicrobial activity) that can account for their claimed probiotic properties. In the present work a comparative proteomic study was performed aimed at characterizing the secretome of four closely related isogenic O/C, SIN, N/R and T B. clausii strains, already marketed in a pharmaceutical mixture as probiotics. Proteomic analyses revealed a high degree of concordance among the four secretomes, although some proteins exhibited considerable variations in their expression level in the four strains. Among these, some proteins with documented activity in the interaction with host cells were identified, such as the glycolytic enzyme enolase, with a putative plasminogen-binding activity, GroEL, a molecular chaperone shown to be able to bind to mucin, and flagellin protein, a structural flagella protein and a putative immunomodulation agent. This study shows, for the first time, differences in the secretome of the OC, SIN, NR and T B. clausii strains. These differences indicate that specific secretome features characterize each of the four strains despite their genotypic similarity. This could confer to the B. clausii strains specific probiotic functions associated with the differentially expressed proteins and indicate that they can cooperate as probiotics as the secretome components of each strain could contribute to the overall activity of a mixed probiotic preparation.